DESIGN AND IMPLEMENTATION OF AN IOT-ENABLED PASSENGER VALIDATION AN AUTOMATED TRAIN STEP DEPLOYMENT SYSTEM

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1. ABSTRACT

The gap between railway platforms and train coaches poses safety risks during passenger boarding. This project presents an IoT-enabled railway safety system that integrates automated step deployment with QR-code-based passenger ticket validation. An AURDINO UNO microcontroller processes distance measurements from an HC-SR04 ultrasonic sensor to identify the train's position and control an MG996R servo motor for step extension and retraction. In parallel, a Wi-Fi-based HTTP interface using ESP32 microcontroller enables smartphone-scanned ticket codes to be checked against predefined validation conditions, allowing authorized access through a relay-controlled gate mechanism. It also counts the no of person on boarding a particular coach. By combining distance sensing, automated mechanical control, and ticket-based access management, the system demonstrates an integrated approach to safer and more efficient railway boarding. The design establishes a foundation for further development of intelligent passenger safety and railway access-control systems.

2. INTRODUCTION

Passenger safety at the platform–train interface remains a critical challenge for railway administration across the world, especially in India. The horizontal and vertical gaps between railway platforms and train coaches, influenced by track curvature, platform geometry, and variations in train positioning, can create hazardous conditions during passenger boarding and alighting. These gaps increase the risk of slips, falls, and foot entrapment. To address these challenges, we have made a mechanical footstep that acts as a platform between coaches and the train.

In parallel, conventional ticket verification systems generally create chaos on a platform, and a lot of people travel without a ticket.

To address these challenges, this project proposes an integrated IoT-enabled railway safety and passenger validation system. An ESP32 and an Arduino UNO microcontroller serve as the central control unit, processing distance measurements from an HC-SR04 ultrasonic sensor to determine whether a train is within a predefined detection range. Based on the programmed conditions, an MG996R servo motor

controls the extension and retraction of a model boarding step. In parallel, a smartphone-based QR-code scanner communicates ticket identifiers to the ESP32 through a Wi-Fi-enabled HTTP interface, allowing the identifiers to be checked against predefined validation conditions and the access gate to be controlled through a relay mechanism. This system also counts the people on boarding the train

By combining sensor-based step control with network-assisted ticket validation and passengers counting system the proposed system demonstrates a unified approach to improving boarding assistance and passenger safety.

3.SYSTEM ARCHITECTURE AND HARDWARE METHODOLOGY:

The proposed system is organised into two functional subsystems: the **IoT-Based Passenger Access and Ticket Authentication Subsystem** and the **Mechatronic Gap-Filler Step Subsystem**. The first manages QR-code-based ticket validation and gate actuation, while the second monitors the train's position relative to the platform and controls the deployment of a retractable step. The ESP32 and Arduino UNO microcontrollers coordinate these operations through sensor inputs, wireless communication, and hardware control signals.

A. System Architecture and Component Integration:

Ticket verification system:

The proposed access-control architecture leverages the ESP32 System-on-Chip (SoC) to seamlessly integrate IEEE 802.11 b/g/n wireless connectivity with real-time mechanical actuation. Configured as a lightweight embedded HTTP server, the microcontroller listens on port 80 for incoming GET requests initiated by optical QR code scans from passenger edge devices. These QR payloads encode a targeted Uniform Resource Identifier (URI) containing query parameters with a unique ticket token. Upon reception, the ESP32's firmware parses the URI payload and executes a validation algorithm against authorized credentials stored in its memory. A successful match yields an HTTP 200 OK response and triggers the hardware-timer-driven Pulse Width Modulation (PWM) peripheral. By offloading the 50 Hz PWM signal generation to dedicated hardware timers, the system prevents network-interrupt-induced CPU jitter, smoothly altering the duty cycle to rotate a servo motor 90 degrees to disengage the physical barrier. The firmware then enforces a strictly timed 3000-millisecond state-holding window to permit passenger ingress before automatically restoring the zero-degree locked position. Conversely, unverified or malformed tokens are systematically

rejected via HTTP 403 Forbidden or 400 Bad Request status codes, ensuring the mechanical barrier remains firmly immobilized against unauthorized access.

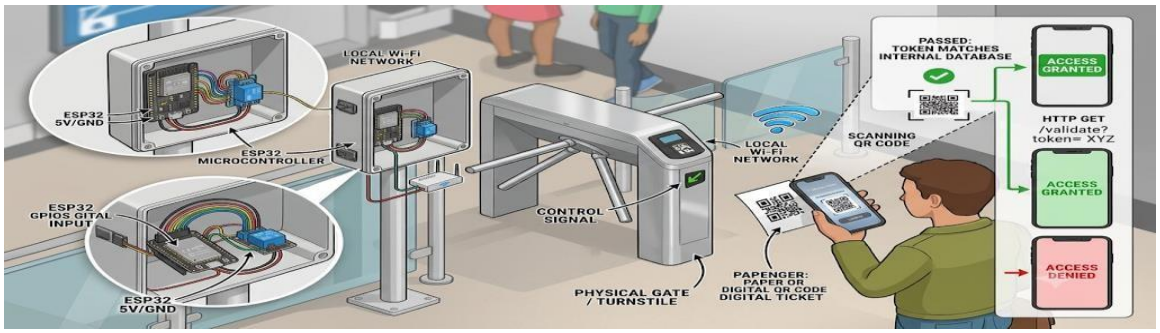


IMAGE (1): QR TICKET SCANNING SYSTEM

Connections:

The hardware integration of the system requires precise electrical connections between the ESP32 microcontroller and the servo motor actuator. The servo motor's positive power terminal (VCC) is connected either directly to the ESP32's 5V/VIN pin for low-current applications or to a dedicated external 5V DC power supply to accommodate higher current demands. To establish a common reference voltage across the system, the servo's ground (GND) terminal is wired directly to one of the ESP32's GND pins; crucially, if an external power source is utilized, its ground must also be tied to the ESP32's ground circuit to prevent signal instability. Finally, the servo's pulse-width modulation (PWM) signal wire is interfaced directly with the ESP32's GPIO 18 pin, providing the dedicated communication channel necessary for the microcontroller to transmit the hardware-timed control pulses required for precise mechanical gate actuation.

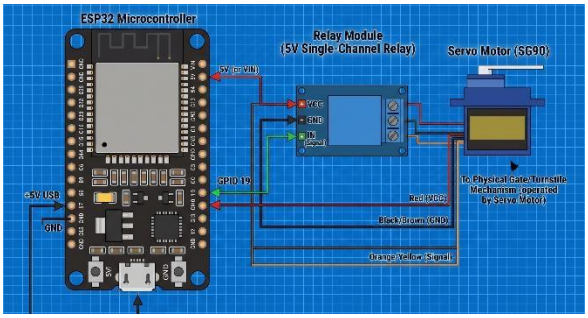


IMAGE (2). CIRCUIT CONNECTIONS

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Train platform interface system

The HC-SR04 sensor measures the distance between the platform and train coach. The Arduino UNO compares this distance with a **distance threshold** to determine whether the step should be deployed or retracted. The MG996R servo controls a hinged step installed beneath the platform. When the measured distance is within the threshold, the servo rotates from approximately **0° to 90°**, extending the step outward. When the distance exceeds the threshold, the step retracts.

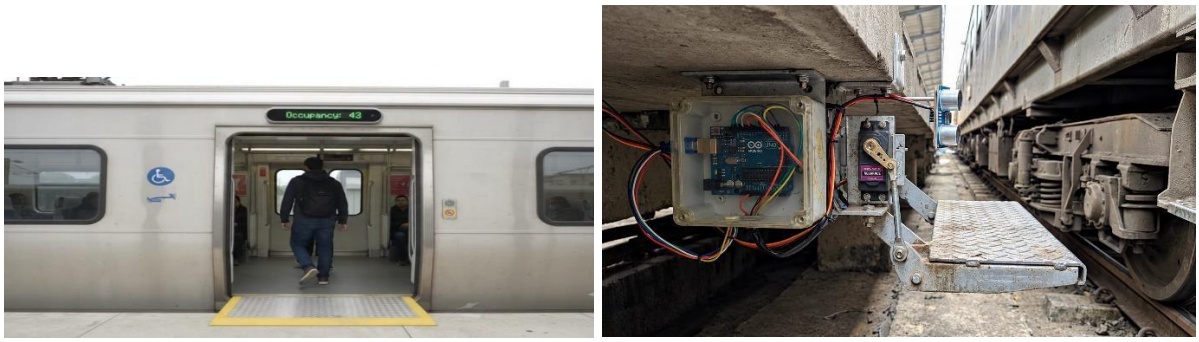


IMAGE (3,4): STEP ATTATCHED TO PLATFORM

4.CONNECTIONS:

The HC-SR04 ultrasonic sensor is integrated using the Arduino's internal 5V logic. The sensor's VCC and GND pins are connected directly to the Arduino's 5V and ground terminals, respectively, while the Trigger (Trig) and Echo data pins are routed to the Arduino's digital pins 9 and 10. The operational status of this sensor can be verified by interfacing the Arduino with a computer and opening the Serial Monitor at a 9600 baud rate, which should yield a continuous stream of real-time distance measurements in centimetres.

The actuation of the bridging step is governed by an MG996R servo motor. Due to its high power requirements, the motor necessitates an isolated power supply while maintaining shared data logic with the microcontroller. The servo's red wire (VCC) is connected to the positive terminal of a dedicated external 5V power supply. Crucially, the ground wire must be connected to both the negative terminal of the external power supply and the Arduino's ground pin; establishing this common ground is essential to prevent erratic vibration and ensure reliable mechanical operation. The signal wire (orange or yellow) is then connected to the Arduino's digital pin 5. Upon system initialisation, the servo automatically calibrates to its retracted state at an angle of zero degrees. The complete subsystem can be mechanically verified by introducing an object within a distance threshold of the ultrasonic sensor, which should prompt the servo to immediately rotate 90 degrees, successfully simulating the dynamic deployment of the platform step.

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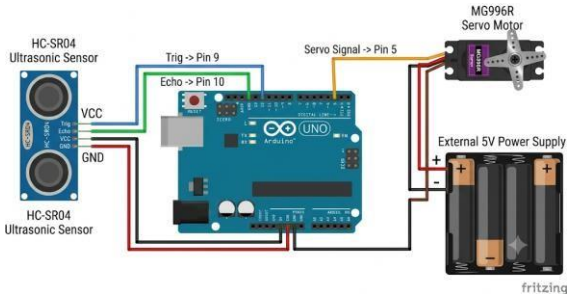


IMAGE (5). CIRCUIT MECHANISM

5.CONCLUSION

This project successfully develops an IoT-enabled system to resolve platform-train gap hazards and unauthorised railway access. Using an Arduino UNO and HC-SR04 ultrasonic sensor, the mechatronic gap-filler dynamically extends a step to prevent slips and falls during boarding. Simultaneously, an ESP32-driven subsystem validates QR-code tickets via Wi-Fi, controlling an automated gate and tracking passenger counts to eliminate ticketless travel and reduce platform congestion.

By integrating real-time sensor data, mechanical extension, and wireless digital authentication, this dual-microcontroller architecture delivers a scalable, secure, and cost-effective foundation for modernizing intelligent railway infrastructure.

6.REFERENCES

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